Note

The condition specified in the if is checked, if it is true, the value of variable1 is stored in variable on the left side of the assignment, otherwise variable2 is taken as the value.

Example 6.4: #Program to check if the accepted number is odd or even (using alternate method of if...else)

```
a = int (input("Enter any number :"))
x="even" if a%2==0 else "odd"
print (a, " is ",x)
```

Output 1:

```
Enter any number :3
3 is odd
```

Output 2:

```
Enter any number :22
22 is even
```

(iii) Nested if..elif...else statement:

When we need to construct a chain of **if** statement(s) then **'elif'** clause can be used instead of **'else'.**

Syntax:

```
if <condition-1>:
    statements-block 1
elif <condition-2>:
    statements-block 2
else:
    statements-block n
```

In the syntax of **if..elif..else** mentioned above, condition-1 is tested if it is true then statements-block1 is executed, otherwise the control checks condition-2, if it is true statements-block2 is executed and even if it fails statements-block n mentioned in **else** part is executed.